Specialty Classification of Clinical Notes: Representations, Metadata and Repeated Notes

Group 14

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1 Introduction

We classify clinical notes into 12 specialties using their description, sample name, transcription and keywords. Data inspection revealed that the first keyword often names the specialty, while identical notes can have different labels. Our research question is: **How do lexical and pretrained representations compare when specialty metadata is available versus absent, and when notes are repeated versus unseen?** We separate model predictions from explicit keyword and duplicate rules to measure these effects.

2 Data

2.1 Data Analysis

Of 2,616 records, three contain note fragments instead of valid labels and have empty clinical fields; excluding them leaves 2,613. Surgery has 876 cases and Dermatology 25 (35:1); 441 cases lack keywords and 30 lack transcriptions. Table 2 gives the class distribution.

Specialty keywords. Of 2,172 keyword lists beginning with a specialty name, 2,171 match the label. Thus metadata can dominate predictions. **Conflicting copies.** There are 712 repeated cleaned-note groups, 711 with conflicting labels; in 710, every copy has a different specialty. For example, "Left lower extremity venous Doppler ultrasound" occurs under both Cardiovascular-Pulmonary and Radiology. **Formatting faults.** Seven descriptions contain embedded records. Shared cleaning truncates them at the first embedded record and normalises the `.,` section separators, identically at fitting and inference.

2.2 Data Augmentation

We target the four smallest classes (Dermatology, Psychiatry-Psychology, Ophthalmology and Neurosurgery), towards 150 cases each, with at most four variants per original. We compare *resampling* (copies), *field dropout* (remove keywords; remove sample name with probability 0.5 and transcription with probability 0.35 when a description remains), and *section recombination*. The latter replaces each headed section, with probability 0.5, with a same-header section from another fitting-fold note of the same class. All originals and donors remain inside the fitting fold. Resampling controls for minority-example frequency.

3 Models

We compare six representation families, informed by the course [Coheur, 2026]. **Naive Bayes (NB)** uses lowercased, Porter-stemmed description TF-IDF with multinomial NB ($\alpha = 1$), matching the specified weak baseline. **Word logistic regression (LR)** uses TF-IDF from all available fields. **Word/character SVM** combines field-specific word and character features. **Latent semantic analysis (LSA) + LR** compresses word TF-IDF into 128 dimensions. **Frozen MiniLM + LR** uses `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019, Wang et al., 2020, Sentence Transformers, 2021]; averaging 220-token chunks yields a 384-dimensional vector per field, concatenated into 1,536 dimensions. **ClinicalBERT** fully fine-tunes `Bio_ClinicalBERT` [Alsentzer et al., 2019]. Since 70.8% of notes exceed 512 tokens, we compare the first 510 word pieces with the first 128 plus last 382, reserving two special tokens. A classifier reads the 768-dimensional [CLS] representation. Character fea-

tures test robustness to medical word fragments and spelling variation; LSA tests whether corpus-learned dense representations improve on sparse lexical features. Frozen MiniLM and fine-tuned ClinicalBERT distinguish general-domain pretraining from clinical adaptation.

Inputs either omit keywords, remove only the first keyword, or retain all keywords. A **keyword rule** takes priority: map the first keyword using fixed specialty names or fitting-data mappings with at least two observations and 95% purity. A **duplicate rule** excludes labels already attached to an identical fitting note and selects the highest remaining model score. This exploits a corpus regularity, rather than a clinical principle.

4 Experimental Setup

We compare stratified *random* partitions, allowing repeated notes across boundaries, with *grouped* partitions, keeping cleaned transcription copies together (other clinical fields identify empty-transcription cases). One seeded outer/inner fold gives 1,672 fitting, 418 validation and 523 held-out cases per regime. Models fit identical rows; selection prioritises validation macro-F1, minimum class F1, then accuracy.

Choices are frozen after validation. In the final pipeline, the official inputs are used only to produce `results.txt`: the winner is refitted on all 2,613 labelled cases and predicts the 398 unlabelled official notes. We report accuracy, macro/per-class F1 and note-group bootstrap intervals.

4.1 Parameters/Hyperparameters

Partition and neural seeds are 20261008 and 42. We search inverse regularisation strength $C \in \{0.3, 1, 3\}$ for SVM and $\{1, 10\}$ for LR, with ordinary or balanced class weights. Larger C weakens regularisation. SVM uses word 1–2-grams (minimum document frequency 2, cap 40,000 per field) and character 3–5-grams within word boundaries (minimum 3, cap 30,000). Word LR/LSA use word 1–2-grams (minimum 2, cap 60,000). TF is sublinear and vectors are L2-normalised. ClinicalBERT uses AdamW, learning rate 3×10^{-5}, weight decay 0.01, 10% warm-up, square-root inverse-frequency class weights, three epochs and effective batch 16 (micro-batches of eight, gradient accumulation two). MiniLM/ClinicalBERT revisions are pinned in the code and references. Runs used an Apple M4 with 16 GB; software includes scikit-learn [Pedregosa et al., 2011], PyTorch, Transformers and Sentence Transformers.

5 Results

Table 1 compares validation-selected bare models, with the first specialty keyword removed (NB uses description only). The submitted system is an **unaugmented SVM**, $C = 0.3$, balanced weights, no keyword input, plus both rules. It achieves **95.79% accuracy, 0.9402 macro-F1 and 0.8000 minimum class F1** on the random held-out set; the 95% macro-F1 interval is [0.897, 0.969]. The separately selected grouped system achieves 93.50% accuracy and 0.9141 macro-F1. These are internal results, not official-test scores. Table 2 gives per-class performance; Figure 1 shows the full confusion matrix.

	Random		Grouped	
Model	Acc.%	F1	Acc.%	F1
NB	36.5	.099	40.7	.143
Word LR	46.7	.502	57.2	.594
Word/char SVM	41.1	.419	59.3	.615
LSA + LR	51.8	.542	54.7	.565
MiniLM + LR	50.1	.542	58.5	.620
ClinicalBERT	53.9	.537	57.2	.604

Table 1: Bare models on common held-out partitions; F1 is macro-averaged. ClinicalBERT uses the validation-selected head+tail variant.

Specialty	N	n	P	R	F1
Cardio-Pulmonary	299	60	.95	.98	.97
Dermatology	25	5	.80	.80	.80
Gastroenterology	192	39	1.00	.90	.95
General Medicine	207	41	.81	.95	.88
Neurology	181	36	.97	.92	.94
Neurosurgery	78	16	.89	1.00	.94
Obstetrics-Gynecology	135	27	1.00	.93	.96
Ophthalmology	71	14	1.00	1.00	1.00
Orthopedic	292	58	.96	.95	.96
Psychiatry-Psychology	40	8	1.00	.88	.93
Radiology	217	44	.96	1.00	.98
Surgery	876	175	.99	.97	.98

Table 2: Submitted system on the random held-out set. N: full labelled-corpus count; n: held-out support; P/R: precision/recall.

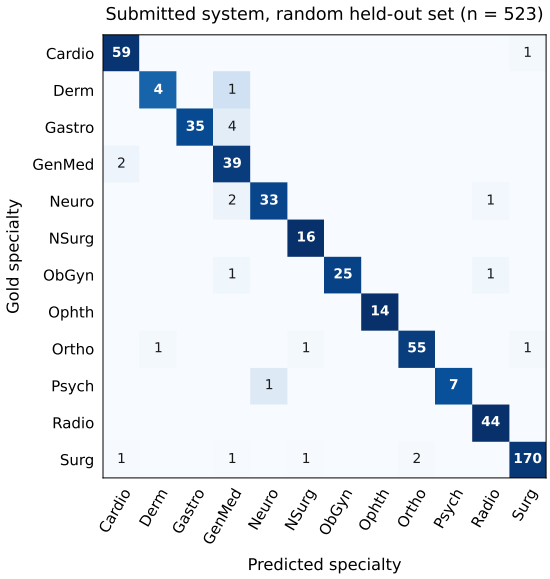

Figure 1: Full confusion matrix of the submitted system on 523 random held-out cases: rows are gold labels, columns predictions; counts are shown, with colour normalised by row. Specialty abbreviations follow Table 2 in the same order.

6 Discussion

Representations and repeated notes. Table 1 shows no clear large pretraining advantage in this single-fold comparison. Of 190 repeated held-out notes, 189 differ in label from their fitting copies. The bare SVM repeats seen labels: 0% accuracy on repeated cases versus 64.6% on unseen cases, demonstrating harmful memorisation.

Metadata and rules. The keyword rule decides 444/523 cases, all correctly; duplicate exclusion decides 25 (76.0% correct), leaving 54 to the SVM (70.4%). Fixed SVM ablations raise random macro-F1 from 0.399 without keywords to 0.885 with all keywords, or 0.945 with both rules (a different configuration from the submission). The gains are corpus-specific.

Augmentation. Table 3 shows only small gains: resampling adds 0.004–0.008 macro-F1, recombination 0.003–0.006; dropout reduces it by 0.011–0.016. Recombination does not outperform simple copies. It can also join incompatible content: an eyelid-repair title from note 1110 accompanies lens-removal text from note 1143.

Errors. The submitted system makes 22 mistakes; the most common pair is Gastroenterology predicted as General Medicine (four). Examples:

Note 361: "I was asked to see the patient for C. diff colitis." Predicted *General Medicine*; gold *Gastroenterology*. With no keywords, the long admission history may obscure the gastrointestinal cue.

Note 2582: "Primary right shoulder arthroscopic rotator cuff repair with subacromial decompression." Predicted *Orthopedic*; gold *Surgery*. The procedure fits both specialties, illustrating ambiguity in a single-label target.

Note 771: "Left third digit numbness and wrist pain." Predicted *Radiology*; gold *Neurology*. A fitting copy is labelled Orthopedic, which duplicate exclusion removes. The sample name "MRI C3 - Cord Compression" favours Radiology, but exclusion cannot identify the correct alternative.

Small class supports, one evaluated fold, a design refined over earlier exploratory runs on the labelled data, and corpus-specific rules limit generalisation; exact matching misses near-copies. Label overlap remains plausible. Lexical and pretrained models perform similarly on unseen notes; metadata and repetition dominate full-system results.

7 Future Work

Repeat outer folds and seeds; test clinically coherent augmentation, whole-note encoding and multi-label targets for overlapping specialties. Evaluate a lexical/neural ensemble only if validation supports it.

Declaration of LLM use. Claude (Anthropic) and OpenAI helped synthesise course material, implement code, perform complex technical tasks and prepare the report. The group made all design and methodological choices and is responsible for the implementation, analysis and conclusions.

References

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Appendix A: Extra Figures and Tables

Regime	Augmentation	Accuracy (%)	Macro-F1	Minority mean F1
Random	None	43.2	.441	.440
	Resampling	43.2	.445	.451
	Field dropout	43.2	.430	.405
	Section recombination	43.0	.444	.450
Grouped	None	59.3	.608	.618
	Resampling	59.5	.616	.638
	Field dropout	58.9	.592	.566
	Section recombination	59.4	.614	.633

Table 3: Augmentation diagnostics for the word/character SVM ($C = 0.3$, balanced weights, first specialty keyword removed, no rules): three-fold pooled predictions on the 2,090 training and validation cases, so the scores are not comparable with the held-out results in Table 1. Minority mean F1 averages the four augmented classes.